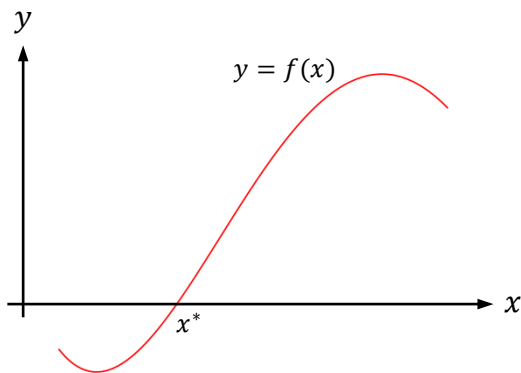


Concept & Outline



Concept:

- 1) Collect all terms to the left-hand side of equation to obtain $f(x)$.
- 2) Select a method to find x 's that make $f(x) = 0$.

② Roots of Equations

- Graphical Method
Guideline for other methods.
 - ✓ ➤ Bisection Method
Robust method.
 - False Position Method
Robust method. (more iterations in some cases)
 - Fixed Point Iterative Method
Slow convergence with rigorous limitation.
 - Newton-Raphson Method
1 initial guess with differentiation.
 - ✓ ➤ Secant Method
2 initial guesses without differentiation.
 - Muller's Method
3 initial guesses, imaginary root.
- Bracketing Method**
An interval bracketing a root is given.
- Open Method**
Initial guesses are not necessary to bracket a root.